

MARCO ANTONIO ADAME RODRÍGUEZ

IoT & Embedded Systems Engineer · Digital Signal Processing · Scientific Software
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PROFESSIONAL SUMMARY

Computer Engineer and graduate of the MSc in Engineering for Innovation and Technological Development, with a terminal specialization in Information and Communication Technologies (UAGro). All coursework completed with a 9.42/10 GPA; degree conferral in progress, expected November 2026. I build end-to-end IoT systems: MEMS sensor, embedded firmware, real-time acquisition, Linux backend, cloud storage and visualization dashboard. My graduate work is a low-cost structural health monitoring (SHM) system I designed from scratch and validated in the field against commercial seismic instrumentation. Lead author of a paper published in Revista Vínculos (Universidad Distrital, Colombia), co-author of two more, and speaker at the 15th International Computing Conference (Tunja, Colombia, 2025). I work daily with scientific Python (NumPy, SciPy, Pandas) and deploy services on Ubuntu Server, Docker and the cloud.

KEY ACHIEVEMENTS

- **A complete SHM-IoT system designed from scratch** and field-validated against the commercial GEA and P-Alert units, at a fraction of the cost of traditional seismic instrumentation.
- **Vibration measurement system on a biaxial shaking table:** synchronized sampling at 200 Hz, FFT analysis over 2048-point windows and 0.98 Hz frequency resolution, with automatic RPM computation from the dominant frequency (published and peer-reviewed).
- **Multi-site monitoring infrastructure running continuously:** nodes on separate networks, a centralized receiver on Ubuntu Server, local buffering against connectivity outages, and published web dashboards.
- **A Python/PyQt5 vibration analysis suite** that processes large volumes of records and generates comparative technical reports automatically.

RESEARCH AND DEVELOPMENT EXPERIENCE

Lead Researcher and Developer — SHM-IoT System · UAGro · MIIDT graduate programme

Aug 2024 – Present

End-to-end design and implementation of a low-cost structural health monitoring solution for civil works, as an accessible alternative to commercial seismic instrumentation.

Acquisition, firmware and hardware

- Integrated acquisition nodes on Arduino Nano RP2040 Connect and Raspberry Pi 4 with MPU9250/MPU6050, LSM6DSOX and ADXL335 MEMS sensors, communicating over I2C, SPI and UART.
- Implemented synchronized acquisition on a dual-core architecture, sampling at up to 200 Hz and satisfying the Nyquist criterion for the structural frequency range of interest.
- Validated the system on a biaxial shaking table, university buildings, a botanical garden, a high school campus and lightweight experimental structures, benchmarking it directly against the commercial GEA and P-Alert units.

Digital signal processing and scientific software

- Built the system DSP pipeline: RMS, FFT, power spectral density (Welch method), dominant frequency, coherence, phase analysis and cross-correlation between sensors.
- Developed desktop applications in Python and PyQt5 for acquisition, live monitoring and analysis, with session control, result export and automatic recovery after failures.
- Built tooling to process TXT, CSV, JSON and JSONL files holding large volumes of records, including time synchronization, resampling, gap detection, digital filtering and series trimming.
- Automated the generation of technical reports and comparative plots documenting system performance against the reference equipment.

Backend, cloud and infrastructure

- Built a centralized receiver that integrates sensors located at different sites and on different networks over HTTP/HTTPS, with multi-origin storage and local buffering that prevents data loss during connectivity outages.
- Developed REST APIs in FastAPI for ingesting, persisting and querying telemetry sent by ESP32 microcontrollers.
- Administered continuously running Ubuntu servers: automatic startup services with systemd, remote access over SSH/VNC, containerization with Docker, and backup and synchronization strategies.
- Published web dashboards and query services that let users visualize, filter and download data from both custom nodes and commercial sensors.

PUBLICATIONS AND TALKS

- **Adame Rodríguez, M. A.**, Alcaraz Vázquez, M., Martínez, E. T., García Guevara, D. S., & Alonso Silverio, G. A. (2026). Ambient vibration measurement system on a biaxial vibrating table. *Revista Vínculos*, 22(1). [Lead author]
- Alcaraz Vázquez, M., **Adame Rodríguez, M. A.**, García Guevara, D. S., Martínez, E. T., & Alonso Silverio, G. A. (2025). Feature-importance-based attribute selection for dengue case classification in Mexico. *Revista Vínculos*, 22(2). [Co-author]

- García Guevara, D. S., Martínez, E. T., Alcaraz Vázquez, M., **Adame Rodríguez, M. A.**, & Alonso Silverio, G. A. (2025). IoT-based biological sample shaker with temperature control. *Revista Vínculos*, 22(2). [Co-author]
- **Talk:** "Ambient vibration measurement system on a biaxial vibrating table". 15th International Computing Conference (CICOM 2025), Tunja, Colombia, October 2025.

TECHNICAL SKILLS

Languages: Python, C/C++ (embedded), JavaScript, SQL, PHP, Bash.

Data and signals: NumPy, Pandas, SciPy, Matplotlib · FFT, PSD/Welch, RMS, digital filters, coherence, phase analysis, resampling and time series.

AI and machine learning: Machine learning, deep learning, computer vision, big data analysis, feature selection.

IoT and hardware: Raspberry Pi, ESP32, Arduino Nano RP2040 Connect · MEMS sensors (IMU/accelerometers) · I2C, UART, SPI, WiFi, Ethernet · real-time acquisition.

Backend and cloud: FastAPI, REST APIs, HTTP/HTTPS, webhooks · Supabase, PostgreSQL, Firebase · distributed synchronization and storage.

Systems and DevOps: Linux (Ubuntu Server/Desktop), Docker, systemd, SSH, VNC, Git/GitHub, Vercel, service automation.

Interfaces and visualization: PyQt5, web dashboards, scientific visualization, export to CSV/JSON/PDF.

SELECTED PROJECTS

SHM-IoT MIIDT — Low-cost structural monitoring · Python, C++, Raspberry Pi, PyQt5

Full architecture for acquisition, storage, analysis and visualization of accelerations in civil structures and experimental models, field-tested against commercial sensors.

Geo-referenced SHM monitoring network · JavaScript, Vercel

Web dashboard showing the location and status of distributed monitoring nodes, continuously deployed and public.

github.com/Ryukert/SHM_GPS_NETWORK_DASH_elite

Supabase sensor dashboard · JavaScript, Supabase, PostgreSQL

Web application for submitting, querying and visualizing measurements stored in the cloud. github.com/Ryukert/MIIDT_SUPABASE_DASH

ESP32 telemetry APIs · Python, FastAPI

REST services for receiving, persisting and querying acceleration data, with scientific plotting tools. github.com/Ryukert/APIAceleraciones

EDUCATION

MSc in Engineering for Innovation and Technological Development · UAGro

Aug 2024 – 2026

Terminal specialization: Information and Communication Technologies (2023 curriculum). **GPA 9.42 / 10** · All coursework completed, no failed courses. Degree conferral in progress.

Thesis: "Development and implementation of a low-cost IoT system for structural health monitoring in civil works".

Courses taken (grades on a 0–10 scale):

- **Artificial intelligence and data:** Machine Learning (10), Deep Learning (10), Computer Vision (10), Data Analysis and Big Data (10).
- **Embedded systems and IoT:** Selected Topics in Embedded Systems (10), IoT Systems Design (9), Fundamentals of Electronic Systems (9).
- **Programme core:** Information and Communication Technologies (10), Sustainable Innovation and Technological Development (10), Graduate Research I (8), II (8) and III (9), Professional Residency and Graduation Project (completed).

BSc in Computer Engineering · UAGro · School of Engineering

2018 – 2023 · Degree awarded 2023

GPA 8.56 / 10 · 55 courses passed. Federal professional license no. 13426767, registered with the Dirección General de Profesiones.

Courses taken by area (grades on a 0–10 scale):

- **Programming and software engineering:** Programming Fundamentals (9), Object-Oriented Programming I (10) and II (8), Advanced Programming (8), Data Structures I (10) and II (9), Computational Logic (8), Systems Analysis and Design (10), Software Engineering (8), Translators and Interpreters (8), Compilers (9), Mobile Application Development (10), Human-Computer Interaction (9).
- **Hardware, electronics and architecture:** Digital Systems (10), Computer Organization (10), Server Architecture (10), Microcontrollers (9), Microprocessors (8), Electrical Circuits (9), Electricity and Magnetism (10), Electronics (7).
- **Networking, operating systems and security:** Routing Fundamentals (10), Operating Systems I (10) and II (8), Network Fundamentals (9), Structured Cabling Systems (9), IT Resource Auditing (9), Network Security (8), Computer Forensics (8), Communications Fundamentals (7).
- **Data, artificial intelligence and signals:** Databases II (9), Databases I (7), Artificial Intelligence Fundamentals (7), Digital Image Processing (7).
- **Mathematics and basic sciences:** Differential and Integral Calculus (8), Probability and Statistics (8), Numerical Methods (8), Analytic Geometry (8), Operations Research (8), Basic Chemistry (8), Vector Calculus (7), Differential Equations (7), Algebra (7), General Physics (7).
- **Research and general education:** Research Seminar I (9) and II (8), Entrepreneurship Workshop (10), Managing Information and Communication Technologies (10), Logical, Heuristic and Creative Thinking (10), Communication Skills (10), Contemporary World Analysis (10), English I (7) and II (7).
- **Practicum:** Professional Internship and Social Service (completed).

ADDITIONAL EXPERIENCE

- **Jóvenes Construyendo el Futuro — Publishing department** · Jun 2023 – Jun 2024. A 12-month vocational training programme in digital production and printing, run by the Mexican Ministry of Labour.
- **Professional internship — IT area, 35th Military Zone (SEDENA)** · Aug 2022 – Jan 2023. 450 certified hours. Computer equipment support and maintenance in an institutional environment with strict operational discipline and handling of sensitive information.
- **Social service — Ignacio Manuel Altamirano Secondary School** · Feb 2022 – Jul 2022. 500 certified hours. Technical support in the computing area and educational support in the community sector.

CERTIFICATIONS AND LANGUAGES

- Data Science Essentials with Python — Cisco Networking Academy / UAGro (Dec 2025).
- Cybersecurity Essentials — Cisco Networking Academy (Mar 2022).
- NDG Linux Unhatched — Cisco Networking Academy (May 2022).
- CCNAv7: Introduction to Networks — Cisco Networking Academy (Jun 2021).
- **Languages:** Spanish native. English intermediate (B1), TOEFL ITP 470 (2025); fluent reading of technical documentation and scientific literature.